Lesson notes · for you

When should it switch?

IoT · Level 3 · Lesson 2

Materials
A clock or a phone stopwatch; Paper and a pen for a table; A device at home that switches itself: a fridge, a water heater or an air conditioner; Optional: a phone for the Mistakes and thresholds lab

Learning goals

- Choose a threshold for a rule from real readings.
- Explain false alarms and misses, and decide which costs more.
- Use two thresholds so a device doesn't flick on and off.

Activity: Find the two thresholds

1. Pick a device at home that switches itself on and off: a fridge's motor hums, a water heater's light glows, an air conditioner's fan changes its sound. Ask first if it is in someone else's room.
2. Draw a table with three columns: time on, time off, and notes (a door was opened, someone used hot water).
3. For an hour, listen or watch, and write the time whenever it switches. Use a clock or a phone stopwatch.
4. Work out each run and each rest in minutes. Are they long and steady, or a few seconds each?
5. Long, steady runs mean two thresholds with a gap; clicking every few seconds would mean one threshold. Write which you think it is, and why. (Some newer “inverter” fridges and air conditioners slow down instead of switching off: if yours never stops, write that down too.)

Don't open any device, touch wiring or hot parts, or unplug anything: just listen and watch. Ask before you go into someone else's room.

Discussion

- Which alarm in your life cries wolf, and what number would you change?
- When is a miss worse than a false alarm?
- Why does a fridge run for minutes, not seconds?

Check yourself

- Give one alarm where a miss costs more, and one where a false alarm costs more.

Try it this week
Find the two thresholds: Find a device at home that switches itself on and off, such as a fridge, a water heater or an air conditioner. Don't open it: just listen, or watch its light. For an hour, write down the time whenever it switches on or off. Then work out how long it runs and how long it rests, and decide whether it uses two thresholds.

Worksheet

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Name _____ Class _____ Date _____

1 **A tank pump switches on below 50% and off above 50%, and it clicks on and off all day. What's the best fix?**

◯ On below 20%, off above 90% ◯ Move the threshold to 60% ◯ Read the tank level less often

2 **To a smoke detector, some cooking steam looks stronger than a small fire. If you raise its threshold until steam never sets it off, no extra fires are missed.**

◯ True ◯ False

3 **Match each alarm to how it should be set.**

A smoke alarm in a dormitory	• • • •	Block only sure spam: hiding real mail costs more
The spam filter on your email	• • • •	Warn early, but not on normal busy days
Your phone's face unlock	• • • •	Go off easily: a missed fire costs lives
The school tank's low-water alarm	• • • •	Open only when very sure: a stranger getting in costs more

4 **A smoke alarm's costs: a missed fire 20, a false alarm 1. Setting A gives 1 miss and 14 false alarms; setting B gives 6 misses and 4 false alarms. Setting A costs less.**

◯ True ◯ False

5 **A greenhouse soil sensor reads 45–60% on watered days, but 15–25% when the drip line is blocked. Choose the alarm's threshold.**

Sound the alarm below _____ %: lower than every normal reading, but higher than every _____ reading.

Word bank: 10 · watered · 35 · blocked

✎ **Think of an alarm or a notification you've learned to ignore. Was its threshold wrong? What would you change it to, and what would that cost?**

Check your answers

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1 **On below 20%, off above 90%**

One threshold flickers wherever you put it, because the level wobbles around the line.

2 **False**

Where the two overlap, a threshold high enough to ignore all the steam also ignores every fire that looks weaker than the steam.

3 **A smoke alarm in a dormitory — Go off easily: a missed fire costs lives**
The spam filter on your email — Block only sure spam: hiding real mail costs more
Your phone's face unlock — Open only when very sure: a stranger getting in costs more
The school tank's low-water alarm — Warn early, but not on normal busy days

Each setting follows from which mistake hurts more.

4 **True**

A: 1 × 20 + 14 × 1 = 34. B: 6 × 20 + 4 × 1 = 124.

5 **Sound the alarm below 35%: lower than every normal reading, but higher than every blocked reading.**

35% sits in the gap between 25% (blocked) and 45% (normal), so normal days stay quiet and a blocked line is caught.